



UNIVERSITÀ DEGLI STUDI DI GENOVA

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DEPARTMENT OF INFORMATICS,
BIOENGINEERING, ROBOTICS AND SYSTEM ENGINEERING

COGNITIVE ARCHITECTURES FOR ROBOTICS

UR5e + OnRobot 2FG7 Integration

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1 Introduction

This report presents the development of a simulated robotic manipulation system composed of a Universal Robots UR5e manipulator and an OnRobot 2FG7 parallel gripper. The system was implemented using ROS 2 Humble, ROS 2 Control, MoveIt2, and RViz. The project focuses on collision-aware pick-and-place planning in a cabinet-like environment. The main implemented features include robot and gripper control, environment modelling through collision objects, inverse kinematics, joint-space interpolation, object attachment and detachment, and random selection of valid placement positions. Since no physical robot was available, grasp validation was performed through a simulated contact-based strategy rather than through real force measurements. The report therefore distinguishes between the implemented simulation-level grasp logic and the force-related capabilities of the real OnRobot 2FG7 gripper, which are discussed through a brief state-of-the-art analysis. The final system executes a complete pick-and-place task. The robot approaches an object placed on a table, closes the gripper using a simulated contact condition, attaches the object to the robot, transports it through a collision-aware environment, places it inside a basket or cabinet area, releases it, and returns safely. The main contribution of the project is the implementation of a deterministic collision-aware motion validation method based on joint-space interpolation.

2 System Architecture

The software stack is based on ROS 2 Humble and includes the following main components:

- **UR5e robot model:** described using URDF/Xacro files;
- **ROS 2 Control:** used to manage simulated arm and gripper controllers;
- **MoveIt2:** used for motion planning, inverse kinematics, forward kinematics, collision checking, and planning-scene management;
- **RViz:** used for visualization of the robot, environment, trajectories, and collision objects;
- **Custom C++ planning node:** implemented in `task_pick_and_place/pick_and_place_task.cpp`.
- **Python launch files:** used to create the easy and hard collision scenes;
- **2FG7 gripper controller:** commanded through a ROS 2 topic.

The UR5e is controlled through the `ur_manipulator_controller`, while the gripper fingers are commanded through the `gripper_position_controller`.

3 Environment Modelling and Collision Avoidance

3.1 Collision Objects

As shown in Figure 3.1, the workspace was modelled using primitive collision objects inserted into the MoveIt Planning Scene. The main objects are the cabinet lower body, cabinet tower, opposite table, placement basket, cylindrical obstacle in the hard scenario, grasped object, represented as `object_box`. All the objects are represented through box primitives and the hard scene contains also an additional cylindrical obstacle, increasing the difficulty of finding collision-free paths.

3.2 Allowed Grasp Contacts

During grasping, contacts between the object and the gripper fingers must be allowed. Otherwise, MoveIt would classify a correct grasp as a collision. A custom allowed-contact logic was therefore implemented. Contacts between `object_box` and the left or right finger pads are considered valid during grasping. All other contacts with the cabinet, basket walls, table, obstacle, or robot links remain forbidden.

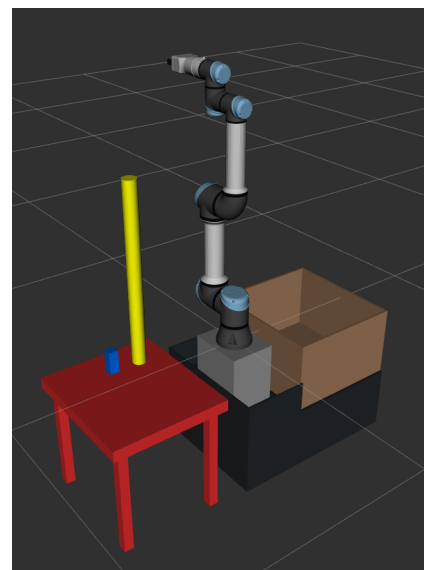


Figure 3.1: Collision objects in the MoveIt Planning Scene.

4 Use of the `moveit_planning` Library

The development of the motion-planning software was based on the open-source ROS 2 package `moveit_planning`[1]. The package provides a C++ wrapper around MoveIt2 functionalities and includes utility methods for inverse kinematics, forward kinematics, trajectory planning, collision checking, planning-scene handling, and object attachment or detachment. These functions were then integrated into the project-specific `task_pick_and_place/pick_and_place_task.cpp` node and combined with custom logic developed for this assignment. Therefore, the `moveit_planning` library provided the general MoveIt2 interface, while the manipulation pipeline and task-specific planning logic were developed and adapted for the UR5e-2FG7 pick-and-place task.

5 Motion Planning and Trajectory Validation

5.1 Joint-Space Interpolation

Instead of relying only on a generic sampling-based planner such as RRTConnect, the project implements a custom joint-space interpolation approach.

Given an initial joint configuration $\mathbf{q}_0$ and a target joint configuration $\mathbf{q}_f$, intermediate states are generated as:

$$\mathbf{q}_i = \mathbf{q}_0 + \frac{i}{N} (\mathbf{q}_f - \mathbf{q}_0), \quad i = 0, \dots, N \quad (1)$$

where N is the number of interpolation steps. For every intermediate configuration, forward kinematics and collision checking are performed. A trajectory is accepted only if all intermediate robot states are valid.

5.2 Inverse Kinematics Validation

For each target end-effector pose, inverse kinematics solutions are computed. A candidate pose is accepted only if: 1) at least one inverse kinematics solution exists 2) the solution is collision-free 3) the interpolated trajectory from the current configuration is valid 4) the attached object does not collide with the environment. This procedure is particularly important for random placement points, since not every geometrically valid point inside the basket is reachable by the robot.

5.3 Cartesian Descent Until Contact

The final placement motion is implemented as an incremental cartesian descent along the vertical direction.

At each step, the end-effector Z coordinate is reduced by a small increment:

$$z_{k+1} = z_k - \Delta z \quad (2)$$

For every new cartesian pose, inverse kinematics and collision validity are checked. The descent stops when the object reaches the intended support surface, the "basket_bottom" in the environment with the bin, or the `cabinet_lower_body` in the environment without it.

6 Pick-and-Place Task Execution

Figure 1 illustrates the complete pick-and-place sequence. In phase **A**, the robot starts from its initial configuration and moves toward a safe pose above the table (**B**). The end-effector approaches the object by reaching a pre-grasp configuration. Phase **C** shows the final motion toward the grasping pose, while **D** provides a close view of the gripper approaching the object. Here the gripper closes and the object is considered grasped; it is then attached to the robot in the MoveIt Planning Scene. The robot subsequently transports the object again in a safe pose above the table **E**. After that, as shown in **F**, the robot moves upon the cabinet and then, in **G**, it reaches the placement area above the cabinet. Phases **H** and **I** show the vertical Cartesian descent until the object reaches the support surface, followed by gripper opening and object detachment. Finally, in phase **J**, the robot performs the reverse Cartesian trajectory and returns safely above the cabinet.

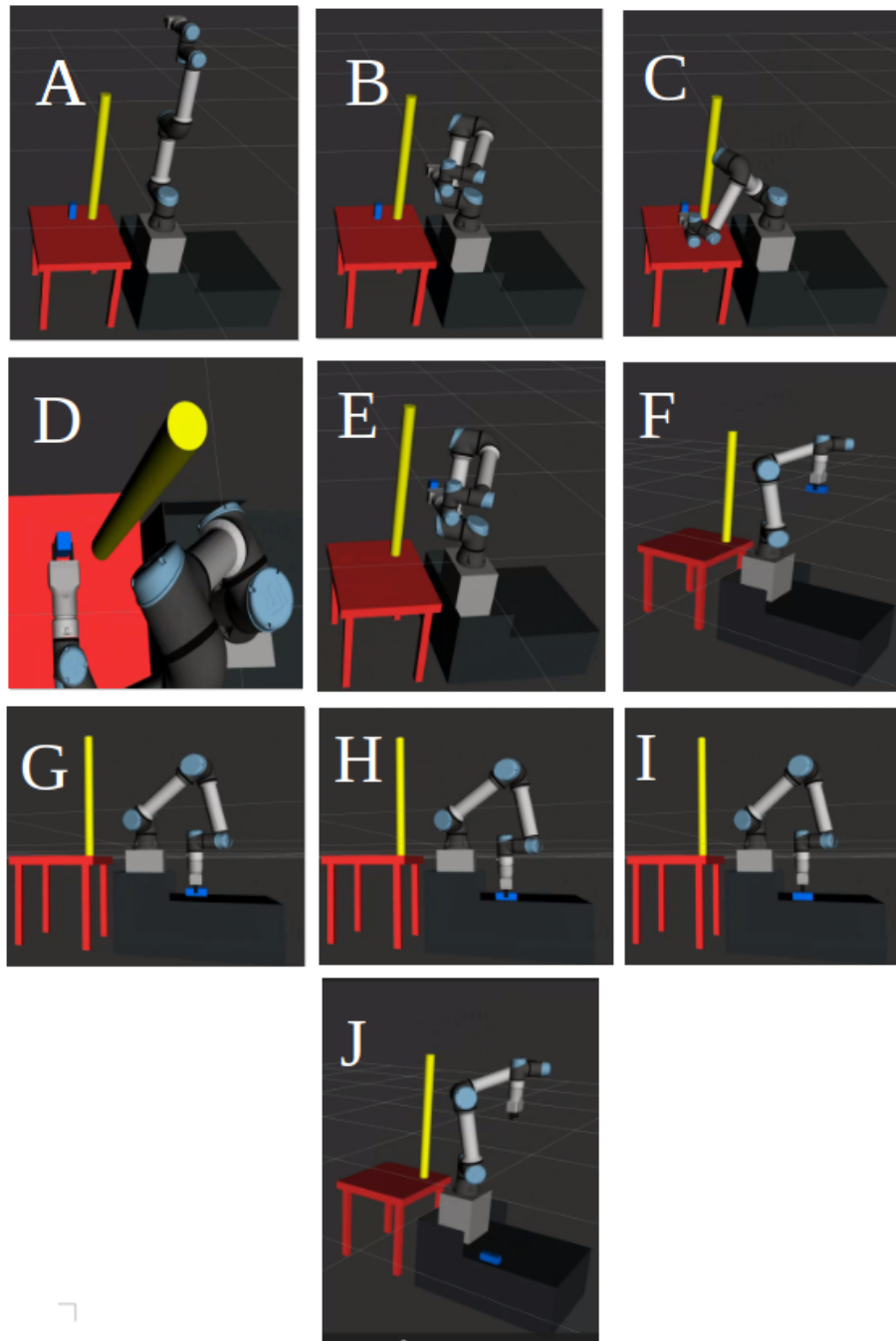


Figure 1: Task's motion sequence

7 Simulated Contact-Based Grasping

The simulated OnRobot 2FG7 gripper is commanded through a ROS 2 position controller. The gripper begins from an open aperture and closes progressively toward a predefined contact aperture. Rather than commanding an instantaneous full closure, the aperture is interpolated in discrete steps.

No real force, torque, current, tactile, or pressure sensor data are used in the simulated setup. Therefore, the grasp is not validated through measured contact force. The object is then attached to `gripper_base_link` in the MoveIt Planning Scene. From that moment onward, the object is treated as rigidly connected to the robot and its collision geometry moves together with the end-effector. This method validates gripper command execution, synchronization between grasp and arm motion, collision-aware transport of the object, and object release and detachment. However, it does not validate real grasp force, contact pressure, object deformation, friction effects, object slip, sensor noise, or force-control stability.

8 Experimental Evaluation

The simulated manipulation pipeline was tested in four configurations: easy scene without basket (A), easy scene with basket (B), hard scene without basket (C), and hard scene with basket (D).

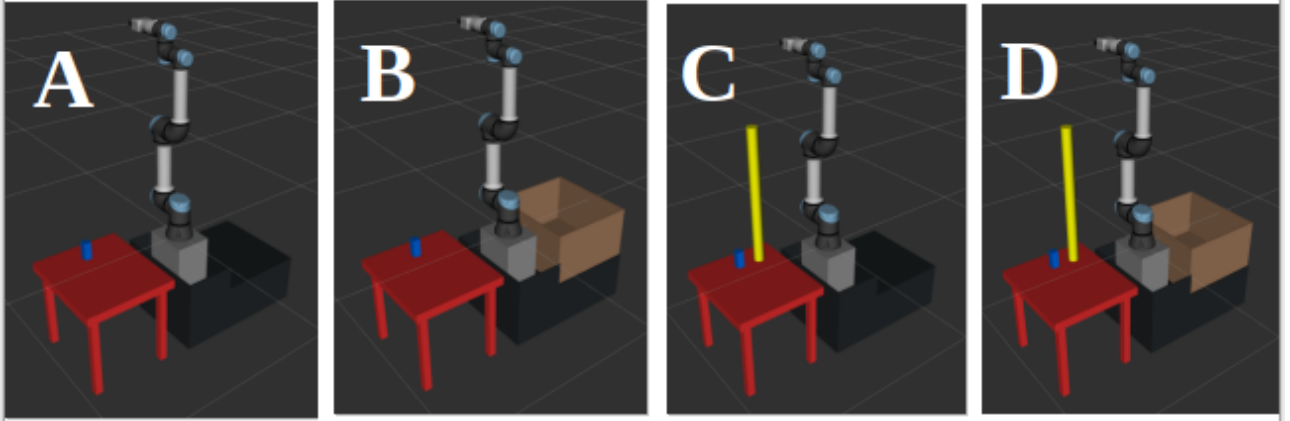


Figure 2: All tested scenarios

One trial was performed for each configuration; therefore, the reported task success rate only indicates that the complete sequence was feasible in the tested run.

Table 1: Quantitative evaluation results for one simulation trial per configuration.

Metric	Easy	Easy + Basket	Hard	Hard + Basket
Number of trials	1	1	1	1
Task success rate [%]	100	100	100	100
Planning time [s]	0.366	0.458	0.542	0.712
Execution time [s]	35.301	44.895	55.891	65.133

The task was completed successfully in all four configurations. Planning time remained below one second in every test, showing that the interpolation-based validation procedure is computationally lightweight.

Execution time increased from 35.301s in the easy scene without the basket to 65.133s in the hard scene with the basket. This is expected because execution time includes all arm motions, gripper opening and closing, and the vertical placement and retreat phases. The hard scene requires longer and more constrained motions, including an additional approach movement before grasping.

9 State of the Art: Force-Related Feedback and Grasp Stability

The OnRobot 2FG7 is an electrically actuated parallel gripper designed for collaborative and industrial manipulation tasks. According to the manufacturer, it provides an adjustable gripping force between 20N and 140N, an external gripping range up to 73mm, and a maximum payload of 11kg [2][3]. These specifications make the gripper suitable for structured pick-and-place operations involving relatively heavy objects and constrained workspaces. Its electric actuation avoids the need for compressed air, while the IP67 protection rating makes the device appropriate for industrial environments where dust or moisture may be present.

Importantly, the official 2FG7 datasheet reports that a force sensor is integrated into the finger located on the connector side of the gripper [3]. This makes the 2FG7 particularly relevant for force-related grasp validation, because contact information can in principle be obtained directly at the gripper rather than inferred only from the commanded aperture. The datasheet also highlights that gravity may affect the force measurement when the workpiece is picked sideways or aligned using the fingers. In such cases, OnRobot recommends orienting the gripper so that the finger containing the force sensor is above the object and ensuring that the lower finger contacts the workpiece slightly before the upper finger. This consideration is relevant for a real-robot implementation, since the measured force would depend not only on the commanded grip but also on gripper orientation and contact geometry.

In robotic manipulation, a grasp cannot be considered successful only because the fingers reached a commanded position. A stable grasp must also maintain the object under external disturbances, such as gravity,

acceleration during transport, contact with the environment, or small modelling errors. Classical grasping studies distinguish between geometric contact, force closure, and grasp stability: the first indicates that the fingers touch the object, whereas the latter requires that the contact forces are sufficient to prevent undesired object motion [4]. In an industrial parallel-gripper application, this means that the gripper should close until contact is detected, apply a sufficient but not excessive gripping force, and maintain that force throughout transport.

Several approaches have been proposed to evaluate grasp stability. In a basic position-based strategy, the gripper aperture is monitored during closure: if the fingers stop before reaching the fully closed position, the system infers that an object is located between them. This approach is simple and widely used in industrial systems, but it does not directly measure contact force and cannot reliably distinguish a stable grasp from a weak or slipping one. A more informative strategy uses motor current, force estimates, tactile sensing, or force-torque measurements at the robot wrist. These signals can be used to detect contact, estimate the applied gripping force, and identify slip or progressive object motion. More recent work has also shown that tactile sensing can detect object rotation or incipient slip and can be integrated into closed-loop regrasping strategies [5].

For a real UR5e-2FG7 system, a practical force-related grasping pipeline would consist of approaching the object, closing the fingers gradually, detecting the first contact through position, current, or force feedback, and then increasing the gripping force only until a stability threshold is reached. After grasping, a short lifting motion could be used as a verification phase: if the object remains stable and no unexpected finger motion, force decrease, or wrist disturbance is detected, the grasp can be accepted. Otherwise, the controller could increase the gripping force or perform a regrasp. This type of closed-loop policy is preferable to a fixed-position closure because it adapts to object size, surface properties, mass, and grasp location.

The current project reproduces only a simplified position-based approximation of this pipeline. The simulated contact aperture represents the finger position at which contact with the object is assumed to occur. In a real implementation, this fixed simulated threshold could be replaced by a force threshold measured by the integrated 2FG7 force sensor. The gripper could stop closing when the measured force exceeds a predefined contact value and then regulate the commanded gripping force according to the object mass, geometry, and required grasp stability.

10 Conclusions

Because the physical UR5e-2FG7 setup was not available, the experimental validation was limited to simulation. The force-related feedback requirement was addressed through the analysis of the official 2FG7 specifications and relevant grasp-stability literature. The developed simulation shows that the proposed UR5e-2FG7 pipeline can perform a complete collision-aware pick-and-place task in a constrained environment. The use of the MoveIt Planning Scene, object attachment, and collision checking makes it possible to verify not only the robot motion, but also the transport of the grasped object inside the cabinet-like workspace.

The custom joint-space interpolation approach proved useful because it provides deterministic trajectories and allows invalid configurations to be identified clearly. However, its main limitation is that it only checks a direct path between the initial and final joint configurations. Therefore, a valid motion may be rejected when an obstacle blocks the direct interpolation, even if an alternative path exists.

A future development would be to compare this approach with RRTConnect. The comparison should include planning time, trajectory length, success rate, number of collision-free solutions, and robustness in both easy and hard scenarios. This would show whether the simplicity and predictability of interpolation are more convenient than the greater flexibility of a sampling-based planner.

Another future step would be to test the system on the real UR5e-2FG7 setup. In particular, the simulated contact threshold could be replaced by the force feedback available from the gripper, allowing grasp detection and grip-force regulation to depend on the actual interaction with the object.

References

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